

LANDSLIDE VISUALIZATION SITUATED ON TANGIBLE TERRAIN MODELS

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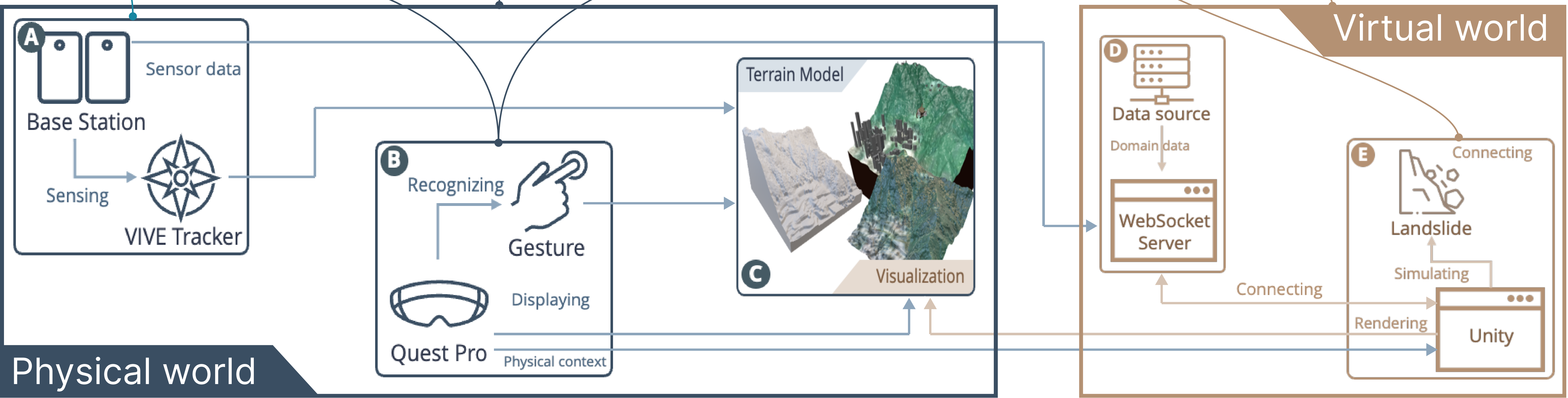
ABSTRACT

Landslides and geohazards are serious threats to public safety and infrastructure, but public awareness is limited. Situated visualizations can engage audiences, but Augmented Reality (AR) interactions can be challenging. In collaboration with experts, we designed LandSAR, an immersive system that combines AR, terrain models, and tangible interfaces. Users can physically interact with terrain models to learn about landslide risks and perceive geographical properties. LandSAR includes an AR-based landslide simulation to educate users about prevention, precautions, and climate change implications. Evaluation results suggest that LandSAR enhances users' experience, making it promising for disaster management, environmental education, and public outreach initiatives.

REQUIREMENTS

- R1 Reduce learning curve**
Considering the target audience is the general public and our primary objective is to promote landslide risk awareness, it is necessary to design a system as a 'walk up and use' model.
- R2 Engage users by novel experience**
The immersive experience is an engaging factor to could draw audiences' curiosity. The proposed system should provide novel experiences and interactions with meaningful messages.
- R3 Foster situation awareness**
Landslides are outside the realm of personal experience. To help users perceive landslides as personal and immediate risks, it is crucial to facilitate situation awareness related to landslides.

LANDSAR



PROTOTYPE

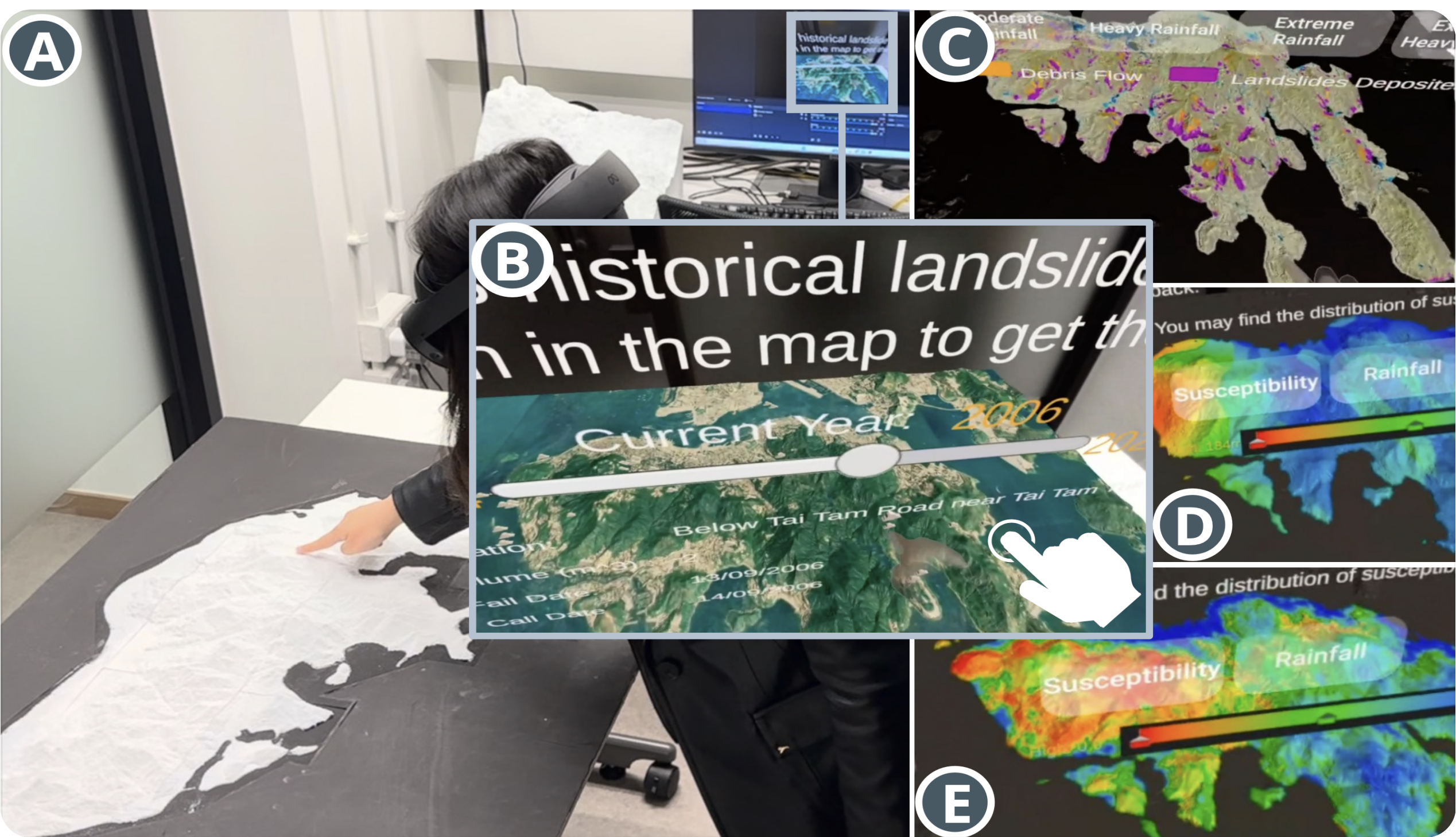


Figure 1. The situated visualization of LandSAR provides historical and predictive information about landslide risks.

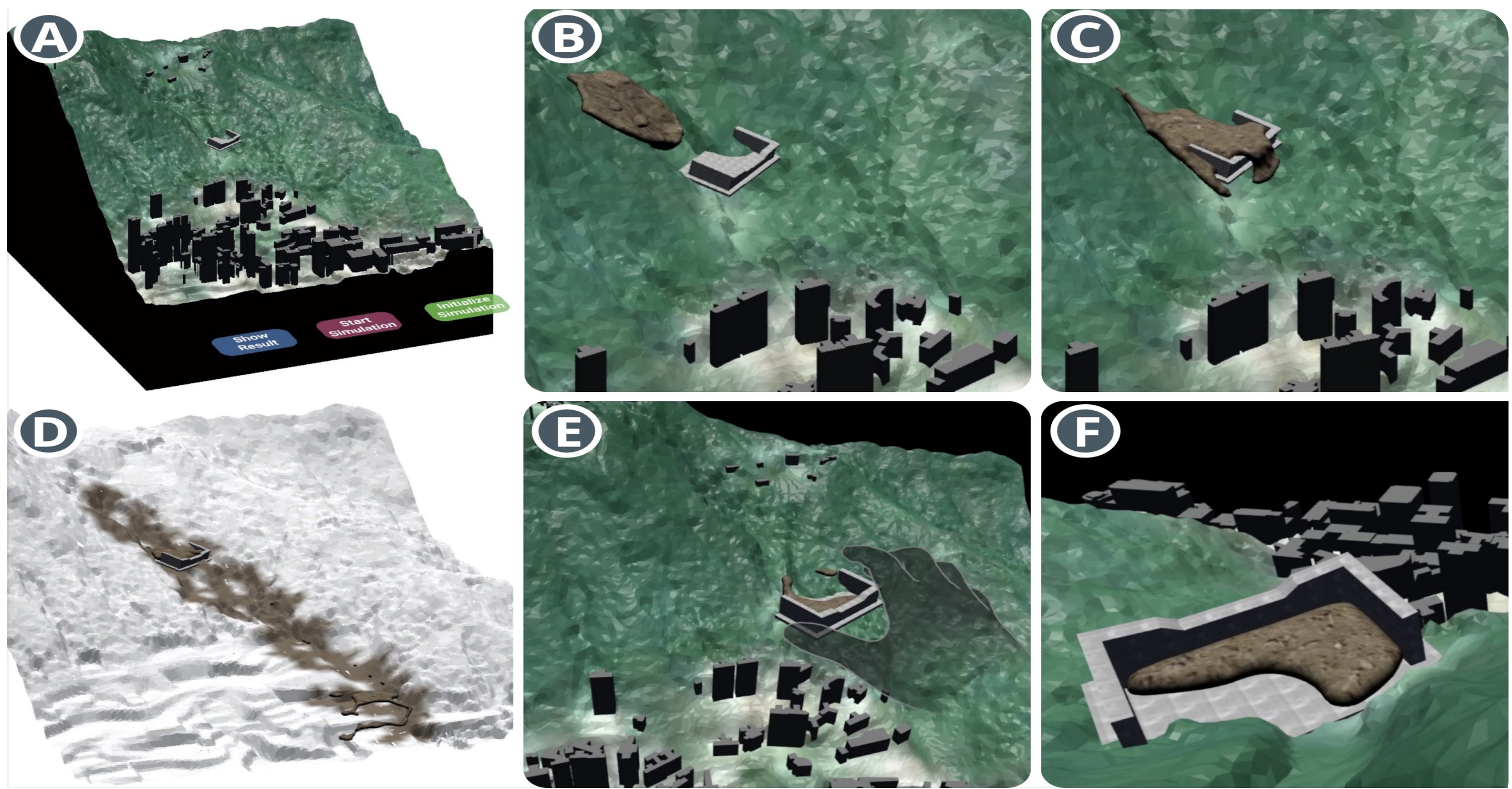


Figure 2. The landslide simulation of LandSAR is designed to enhance situation awareness.

EVALUATION

Obstructive	1	6	5	Supportive		
Complicated	1	1	3	5	2	Easy
Inefficient	2	7	3	Efficient		
Confusing	5	5	2	Clear		
Boring	3	4	5	Exciting		
Not interesting	1	5	6	Interesting		
Conventional	3	6	3	Inventive		
Usual	1	7	4	Leading edge		

1 2 3 4 5 6 7

Figure 3. The User Experience Questionnaire (UEQ). The color indicates the marks. The numbers indicate the number of users.

DISCUSSION

- Contribution:**
- LandSAR: an immersive system prototype designed to enhance situational awareness of landslide risks.
 - LandSAR provides users with a seamless and intuitive tangible interface for physically interacting with landslide simulations.
 - Our work proves the concept of using terrain models to bridge the gap between the physical and virtual worlds.
- Limitation:**
- The discrepancy between simulation and actual mechanisms, which highlights the need for exploring improved simulation techniques.
 - Touching on the terrain models is tracked by Oculus, which requires substantial attention to the hand positions. However, this is not ideal if users want to fully immerse or collaborate with others. Thus, external-tracking systems might be needed.